#78

Examining the impact of AI in open source software projects

AI-Use Disclosure Governance and Contributor Compliance in Open-Source Software

Background

AI-assisted development is increasingly used in open-source software (OSS). While AI tools can improve developer productivity, they also raise **concerns about transparency, quality, licensing, and accountability**.

Some OSS projects have responded by introducing **policies requiring contributors to disclose AI use.**

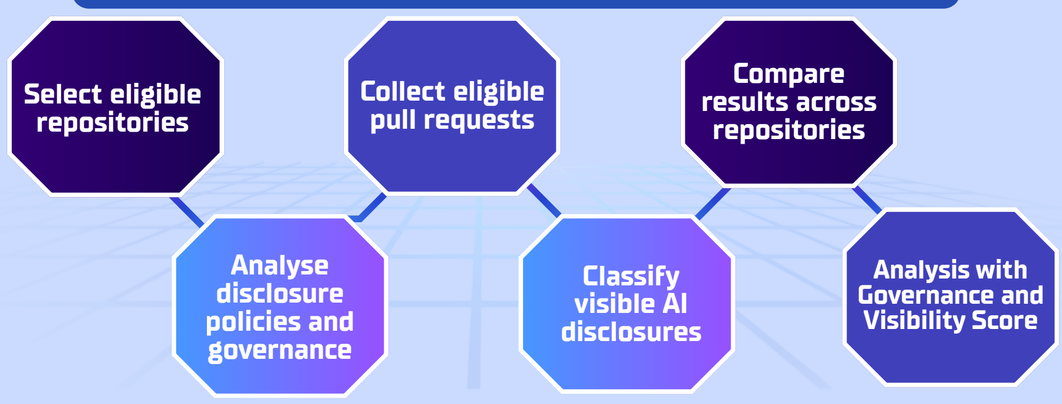
Questions

RQ1.
How are OSS projects **adapting** to AI-assisted development?

RQ2.
How well do contributors **comply** with AI disclosure policies?

RQ3.
How do policy **visibility and governance influence** disclosure rates?

Methodology



Approach: Quantitative PR analysis and qualitative policy analysis

Dataset

41
Open Source Repository

41,000
Pull Requests

Repository Criteria

- Public & OSS-licensed
- Active (>20 commits/year)
- Popular (>500 stars)
- English documentation
- PR workflow & AI policy

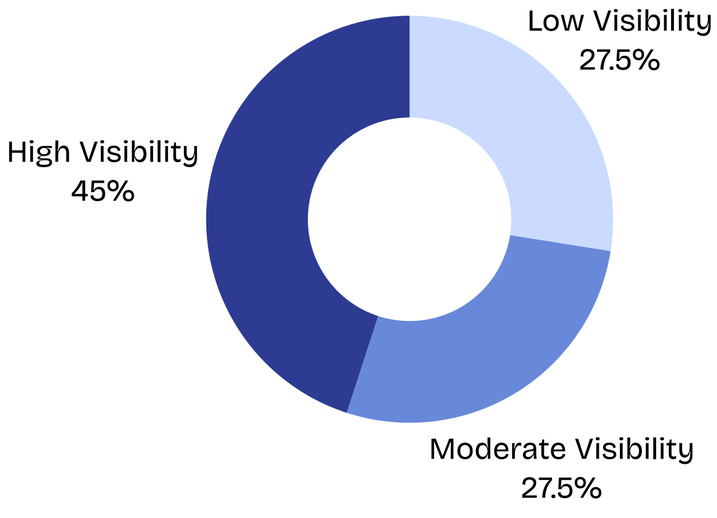
PR Criteria

- Closed PRs
- Human contributors only
- Latest 100 PRs per repo
- 4-month observation window
- Duplicates & incomplete data excluded

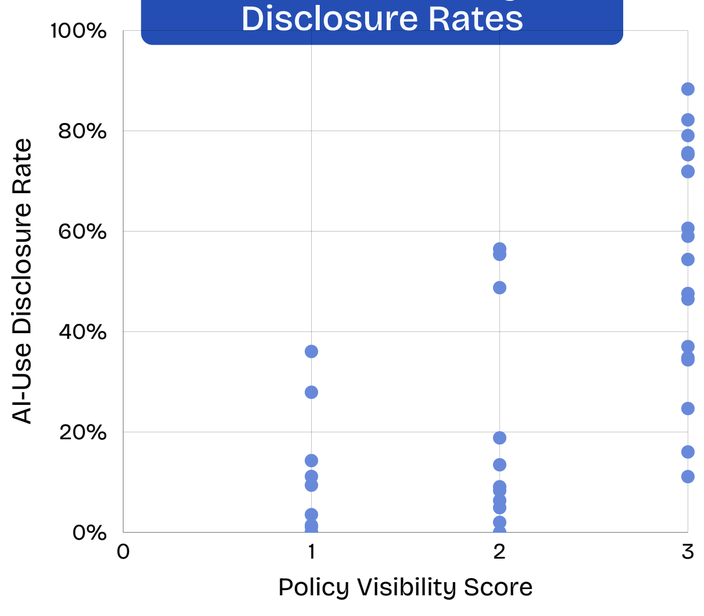
Analysis/Result

Across 41,468 analysed PRs, **27.95%** (11,592) contained AI-use disclosures, with higher disclosure rates associated with greater policy visibility. Variation in policy visibility and stated consequences highlights inconsistencies in how OSS projects communicate and enforce AI-use disclosure requirements.

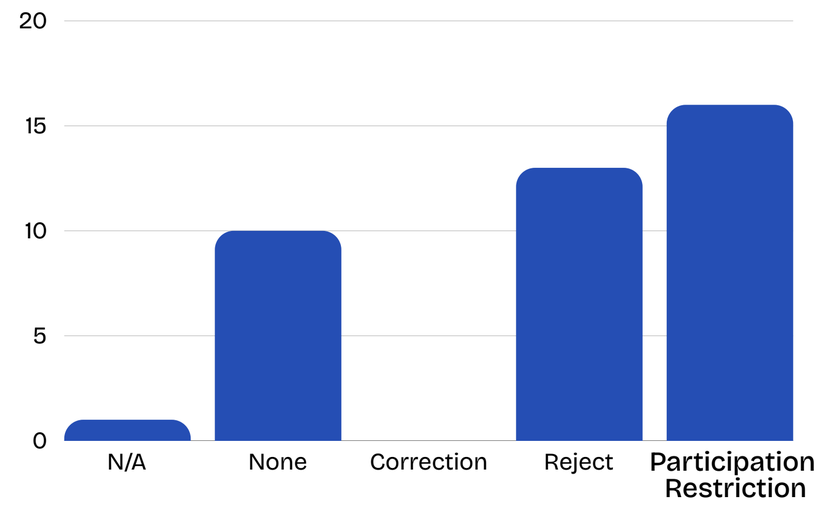
Policy Visibility Distribution



AI Disclosure Policy v.s Disclosure Rates



Consequences Stated



Conclusion

Higher policy visibility was associated with greater disclosure rates, suggesting that integrating clear requirements into contribution workflows may improve compliance. Inconsistent enforcement mechanisms highlight the need for more transparent and standardised AI governance practices across OSS projects.

Anna Chibel · Minseo Kim

Supervisor: Kelly Blincoe